

$$0.4142135623730951 = \frac{1}{b + \frac{1}{c + \frac{1}{d + \frac{1}{e + \frac{1}{f + \frac{1}{g + \dots}}}}}}$$

Again, we can take a reciprocal of this:

$$\frac{1}{0.4142135623730951} = 2.4142135623730945 = b + \frac{1}{c + \frac{1}{d + \frac{1}{e + \frac{1}{f + \frac{1}{g + \dots}}}}}$$

Our next term, b , will be the integer part to the left of the decimal point in this new term—in this case, 2. And then we will repeat the process: taking a reciprocal of a decimal part, finding the integer part to the left of the decimal, and so on.

In Python, we accomplish each round of this as follows:

```
next_term = math.floor(1/leftover)
leftover = 1/leftover - next_term
output.append(next_term)
```

We can put the whole process together into one function as in [Listing 5-6](#).

```
def continued_fraction_decimal(x,error_tolerance,length_tolerance):
    output = []
    first_term = int(x)
    leftover = x - int(x)
    output.append(first_term)
    error = leftover
    while error > error_tolerance and len(output) <length_tolerance:
        next_term = math.floor(1/leftover)
```
